

$$A = 4R^2 \cos\theta \sin\varphi$$

$$\nabla A = 8R \cos\theta \sin\varphi \hat{r} - 4R \sin\theta \sin\varphi \hat{\theta} + 4R \cot\theta \cos\varphi \hat{\varphi}$$

$$\nabla \cdot \nabla A = 24 \cos\theta \sin\varphi - 8 \cos\theta \sin\varphi - 4 \frac{\cos\theta}{\sin\theta} \sin\varphi$$

$$\nabla \cdot \vec{B} = 0 + \frac{\cot\theta \sin\varphi}{R}$$

برای مختصات کروی

$$\nabla A = \frac{\partial A}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial A}{\partial \theta} \hat{\theta} + \frac{1}{r \sin\theta} \frac{\partial A}{\partial \varphi}$$

$$\begin{aligned} \nabla \cdot \vec{B} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 B_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta B_\theta) \\ &\quad + \frac{1}{r \sin\theta} \frac{\partial B_\varphi}{\partial \varphi} \end{aligned}$$

(الف)

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V \nabla \cdot \vec{F} d\tau$$

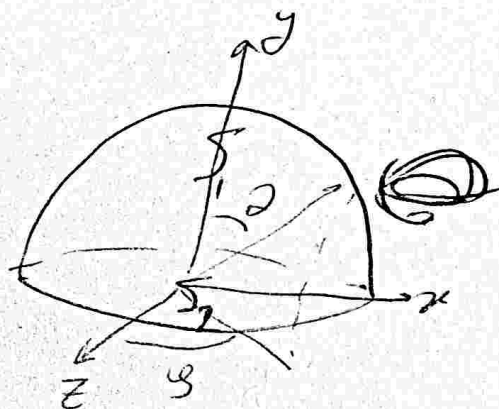
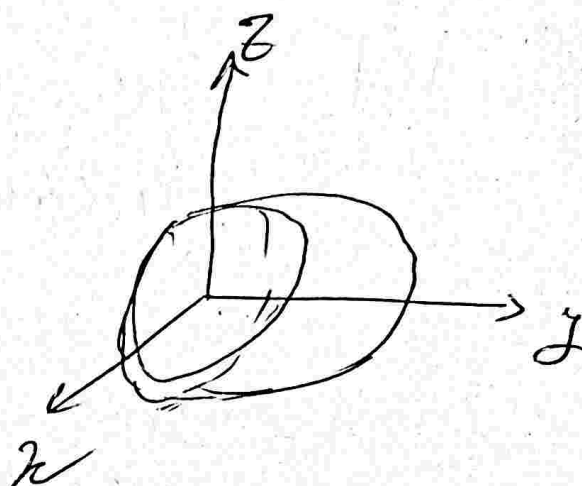
$$\nabla \cdot \vec{F} = 1$$

$$\oint_S \vec{F} \cdot d\vec{S} = \frac{2}{3} \pi R^3$$

$$n_1 = \frac{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}{R} \quad n_2 = -\hat{y}$$

$$\int_S \vec{F} \cdot d\vec{S} = \int_{S_1} \vec{F} \cdot d\vec{S}$$

$$= \int_{S_1} \frac{4x + x^2}{R} dS$$



$$, \quad x = R \sin \theta \cos \phi$$

$$\bar{I}_1 = 4R^2 \int_0^{\pi/2} \frac{\sin \theta \sin \varphi \sin \theta d\theta d\varphi}{r/R} = 0$$

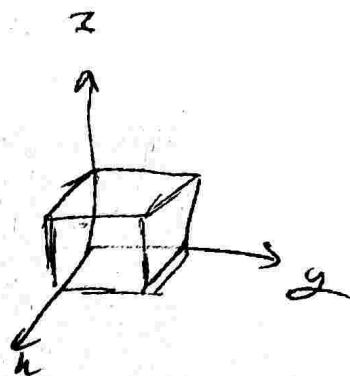
$$\bar{I}_2 = R^3 \int \sin^2 \theta \sin^2 \varphi \sin \theta d\theta d\varphi$$

$$= 1\pi R^3 \int_0^{\pi/2} \sin^3 \theta d\theta = \frac{2}{3} \pi R^3$$

$$\int_S \vec{F} \cdot d\vec{S} = \bar{I}_1 + \bar{I}_2$$

همانطور که انتظار داریم دو انتگرال با هم برابر هستند و البته $\nabla \cdot \vec{F}$ بسیار راحت قابل محاسبه است.

$$\oint_S \vec{F} \cdot d\vec{S} \stackrel{!}{=} \int_V \nabla \cdot \vec{F} \, dV$$



$\hat{x}$:

$$\vec{F} \cdot \hat{x} = x^2$$

$$\oint_{S_1} \vec{F}(\hat{x}) \, dS = - \int_0^1 x^2 \, dx = -1/3$$

$$\oint_{S_2} \vec{F} \cdot \hat{x} \, dS = 1/3$$

$\hat{y}$:

$$\oint_{S_3} (2x + z^2) \, dx \, dz = -1/3$$

$$\oint_{S_4} (2x + z^2) \, dx \, dz = 1 + 1/3 = 4/3$$

$\hat{z}$:

$$\oint_{S_5} (2xz) = 0 \quad \Rightarrow \quad \oint \vec{F} \cdot d\vec{S} = 2$$

$$\oint_{S_6} 2x \, dy \, dz = 1$$

$$\oint \nabla \cdot \mathbf{F} d\tau$$

$$= \int (2x + 2y) dx dy dz = 1 + 1 = 2$$

$$\nabla \times F = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{1}{r} \right) \hat{z} = 0 \quad r \neq 0$$

$$\oint_C \vec{F} \cdot d\vec{L} = \oint \frac{1}{a} a d\theta = \frac{2\pi}{\cancel{a}}$$

$2\pi \neq 0$ به ظاهر قضیه استوکس برقرار نیست! به این خطرات که
 در نقطه‌ای $r=0$ (یعنی z محور) $\nabla \times F$ تعریف نشده است و این عامل مشکل است

$$\nabla \times F = \cancel{2\pi} \delta_r \hat{z} \quad \text{اگر فرض کنیم}$$

$$\iint \delta_r ds = 2\pi \int \delta_r dr = 2\pi$$

قضیه استوکس برقرار می‌شود.